

DRMD Best Practice (v1.0)

16.2 Software/LIMS Checklist

A. Parser Configuration & Namespace Handling

- ☐ Namespace Awareness: Parser matches elements by URI + local name.
- ☐ Prefix Independence: Logic MUST NOT rely on document-provided prefixes.
- ☐ URI Mapping: Stable internal map for DRMD, DCC, SI, XMLDSig.
- ☐ XXE Protection: External entity resolution MUST be disabled.
- ☐ Schema Resolution: Resolve imports from trusted, pinned offline bundles.

B. Validation & Conformance Layers

- ☐ Structural Check: Perform XSD validation against drmd.xsd and all imported schemas.
- ☐ Profile Determination: Automatically classify the document as the highest profile (A, B, or C).
- ☐ Business Rule Enforcement: Verify mandatory narratives, finite numbers, uncertainty boundaries.
- ☐ D-SI Unit Validation: Implement regex checks for unit strings (e.g., flagging double prefixes).
- ☐ Reporting: Generate a machine-readable JSON report.

C. Data Extraction & Normalization

- ☐ Unified Quantity Algorithm: Implement a single extraction logic prioritizing si:real.
- ☐ Multilingual Selection: Implement the fallback algorithm (requested lang -> no lang -> first block).
- ☐ Identifier Keys: Normalize identifiers by trimming whitespace and using (scheme, value) pair.
- ☐ Unit Preservation: Store original D-SI unit string as-is.
- ☐ Numeric Precision: Store values in high-precision types (double/decimal) without rounding.

D. Security & Trust Management

- ☐ Cryptographic Verification: For Profile C, use full XMLDSig library.
- ☐ Attachment Safety: Enforce size limits, verify MIME types, scan for malware.
- ☐ Version Handling: Inspect schemaVersion and implement strategy-based parsing.